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Post-fire regeneration of endangered limber pine (*Pinus flexilis*) at the northern extent of its range

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ABSTRACT

Limber pine (*Pinus flexilis*), an understudied tree species important to montane and subalpine ecosystems, is listed as endangered in Alberta. Dispersal of seeds to newly disturbed, open areas by Clark's nutcracker (*Nucifraga columbiana*) is expected to facilitate post-disturbance establishment of limber pine. Prescribed burning has thus been proposed as a tool to stimulate natural regeneration, but no studies have yet surveyed limber pine's regenerative response to disturbance at the northernmost extent of its range. We examined post-fire regeneration in one prescribed burn and one wildfire at the northern edge of limber pine's distribution and compared this to a regeneration baseline of plots in nearby unburned limber pine acting as seed sources to the burns. Using a likelihood-based approach, we then compared hypotheses of ecological processes influencing seedling occurrence and abundance. Overall, we found only six post-fire limber pine seedlings within the burns, as compared to one hundred twenty-four similarly aged seedlings found in unburned plots. Akaike information criterion (AIC) identified distance to a seed source and absence of canopy cover as hurdles for seedling establishment in the burn. Analysis of seedling abundance was performed on only the unburned dataset due to low regeneration in the burned areas surveyed; model selection here showed that when distance to a seed source is small, substrate variables, including less prevalent bare mineral soil, more rocky cover, and finer soil textures, take on an important role in driving abundance. Substrates important for limber pine abundance in unburned stands were found with similar frequency in the burns, indicating limber pine colonization post-disturbance was not substrate limited but instead constrained by other factors. Our findings suggest that extensive prescribed burns and stand-replacing wildfire events at the northern extent of limber pine's range may have limited seedling regeneration in the immediate post-fire period. Conservation efforts may be well-served by focusing on fire mitigation efforts, such as thinning and other fuel treatments, in areas surrounding established limber pine stands. Finally, post-disturbance stands could benefit from supplementary seedling plantings to achieve recovery plan restoration goals for this endangered species.

1. Introduction

Limber pine (*Pinus flexilis*), an understudied tree species crucial to ecosystem function in the montane and subalpine regions of Alberta, was listed as *Endangered* in Alberta in 2008 (Alberta Whitebark and Limber Pine Recovery Team, 2014), and assessed as *Endangered* by the Committee on the Status of Endangered Wildlife in Canada (COSEWIC, 2014). It is most strongly threatened by the rapid increase in mortality and decrease in recruitment generated by the introduction of white pine blister rust (WPBR) (Kinloch, 2003). This often-lethal disease, caused by the fungus *Cronartium ribicola*, has spread through the population, affecting an estimated 88% of Alberta's limber pine stands (Smith et al., 2013). Although some individuals are known to harbour rust-resistance (Schoettle et al., 2014), outbreaks of mountain pine beetle (*Dendroctonus ponderosae*), driven by warming climate, have the potential to attack and kill these stands of high-conservation value (Langor, 1989), further reducing resilience to the disease. Loss of limber pine from these landscapes will result in a diminishing food and habitat source for many species (Schoettle, 2004) along with decreased slope stability and moisture retention (Baumeister and Callaway, 2006; Tomback and Achuff, 2010) on the wind-swept, steep, rocky sites it calls home.

Fire, although certainly a threat to limber pine forests, has been speculated to also play a role in promoting limber pine regeneration. This is due to limber pine's mutualistic relationship with the Clark's

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nutcracker (*Nucifraga columbiana*), a bird which scatterhoards limber pine seed across the landscape (Lanner and Vander Wall, 1980). Some of these caches are buried 1–2 cm in the ground, from which unclaimed seed, if placed in safe sites suitable for growth, can subsequently germinate (Hutchins and Lanner, 1982; Tomback, 1982). Clark's nutcrackers have been observed caching in open areas, such as those created by fire, perhaps to capitalize on the enhanced rate of snowmelt or removal of snow by wind that provides earlier and easier access to cached seed (Vander Wall and Balda, 1977). It is thus thought that fire may promote nutcracker caching in open areas in which limber pine can germinate, free from suppression from competitor species (Donnegan and Rebertus, 1999; Tomback and Linhart, 1990).

Limited studies, however, have been conducted on the direct relationship between limber pine regeneration and fire. Of these, most have been dendrochronological studies which have shown limber pine as the first species to establish following wildfire events, citing nutcrackers as providing a competitive advantage in long-distance dispersal to extensive burns or in recolonizing limber pine stands extirpated by fire (Donnegan and Rebertus, 1999; Rebertus et al., 1991; Webster and Johnson, 2000). One survey of natural regeneration following fire has been conducted in the Colorado Front Range; in this, the authors found regeneration within burned areas, though the relationship between regeneration density and burn severity varied among the three fires surveyed (Coop and Schoettle, 2009).

Currently, much of the research on fire and five-needle pines has been on whitebark pine (*Pinus albicaulis*), a species similar to limber pine in regard to its reliance on Clark's nutcracker for seed dispersal. The two species differ in that limber pine cones open when mature, and thus some seeds may be dispersed by other birds or fall to the ground and be dispersed by rodents, though this contribution to regeneration is very minor (COSEWIC, 2014). Several whitebark pine studies have documented a fire-mediated boost in either seedling density, growth, or survival in both natural regeneration surveys (Retzlaff et al., 2018; Tomback et al., 2001, 1993) and seedling planting trials in burned soil (Cripps et al., 2018; Lonergan et al., 2014; Perkins, 2015). However, many recent studies of whitebark pine are revealing that the relationship between five-needle pines and fire may be more nuanced than once thought. The absence of competition may be the key component to promoting whitebark regeneration rather than fire itself (Cripps et al., 2018), with opening of the canopy only promoting growth release of seedlings in more closed canopy whitebark stands (Maher et al., 2018)-findings likely applicable to competition-intolerant limber pine (Peters and Visscher, 2019). Nutcracker caching is also proving to be based upon complex decision-making, with findings of nutcrackers caching in many forest types (Goeking and Izlar, 2018) and even avoiding openings, preferentially caching in treed areas (Lorenz et al., 2011). There is growing acceptance that fire management for whitebark forests differs among regions, demanding consideration of stand characteristics, such as openness and presence of other tree species (Keane, 2018).

Results from studies of whitebark pine that have found a positive relationship between recruitment and fire, along with the findings from the limited number of limber pine and fire studies, have driven proposals of using prescribed burning to remove competition and stimulate natural regeneration of limber pine (Alberta Whitebark and Limber Pine Recovery Team, 2014). It is important to note, however, that important ecological differences exist both between whitebark and limber pine, as well as among geographically disparate populations of limber pine in which studies to date have been conducted. These differences, including elevation range and resulting effects on overall stand composition, co-occurring vegetation, and climate (Coop and Schoettle, 2011; Steele, 1990), may have important impacts on nutcracker caching decisions, fire behaviour, and the resulting regenerative response to disturbance. Although management decisions must always be made in the presence of uncertainty, the scarcity of studies on limber pine and more recent acknowledgment of spatially varying responses to fire in the far better studied whitebark pine, highlight the need for a focused study of limber pine and disturbance to allow more informed use of prescribed burning as a recovery tool.

This study represents a first look at immediate post-fire regeneration of limber pine at the northern limit of its distribution. Specifically, we examined limber pine regeneration on one prescribed burn and one wildfire that occurred within dispersal distance of limber pine stands 8- and 16-years post-fire. Both burns were large, allowing us to test whether extensive burns provide a competitive advantage to limber pine in this context, as has been suggested by studies in other regions (Coop and Schoettle, 2009; Donnegan and Rebertus, 1999; Lanner and Vander Wall, 1980). Our objectives were to determine:

1. How does limber pine recruitment vary between burned and unburned areas? Within burned areas, does recruitment differ between stands which were limber pine prior to disturbance versus those without limber pine where disturbance could promote opportunities for new colonization?
2. How do factors such as seed availability and dispersal, substrate composition, microclimate, or presence of competing vegetation promote or limit natural limber pine regeneration occurrence and abundance?
3. How do the variables driving seedling abundance compare between burned and unburned areas?

Increased understanding of the response to and mechanisms driving post-fire regeneration of limber pine is crucial for proper management of this ecologically important resource. By answering the questions above, we aim to help fill the knowledge gap concerning the role of fire in limber pine recruitment and inform the use of fire in recovery planning for this endangered species.

2. Material and methods

2.1. Study areas

We surveyed limber pine regeneration within two recently burned areas and surrounding seed source stands at the northernmost extent of its range in Alberta's R11 Forest Management Unit (FMU) (Fig. 1). In 2017, the 2001 Dogrib wildfire (51.67° N 115.42°W, ~10,000 ha), and the 2009 Upper Saskatchewan prescribed burn (52.03°N 116.60°W, ~5600 ha) were selected for study on the basis of their size and proximity to mature limber pine that could act as a seed source for regeneration to the burn. Prior to fire, the area burned by Dogrib had been coniferous forest dominated by white spruce (*Picea glauca*) and lodgepole pine (*Pinus contorta*), with scattered small stands of limber pine interspersed throughout rockier, less hospitable regions (Table 1). The Dogrib fire perimeter contained several small limber pine patches (4.4–8.8 ha) that escaped crown-fire, with a few similarly sized limber pine stands present outside of the burn perimeter, within known nutcracker dispersal distance. The interior of the Upper Saskatchewan fire had been predominately lodgepole with a minor component of spruce, with limber pine stands occurring along ridges, at cliff edges, and along the banks of the nearby North Saskatchewan River (Table 1). No substantive limber pine was found within the fire perimeter that could act as a seed source; instead, a large limber pine stand (~388 ha) bordering the east side of the fire perimeter, along with a few small stands to the south and southeast along the North Saskatchewan River, acted as the major potential seed sources. Sites surveyed were placed in more severe portions of the two burns, as described by overstory mortality (Table 1). In the Dogrib region, sites were primarily in the montane, whereas Upper Saskatchewan sites included a mix of montane and subalpine areas.

2.2. Field sampling

Limber pine regeneration was sampled at three stand types

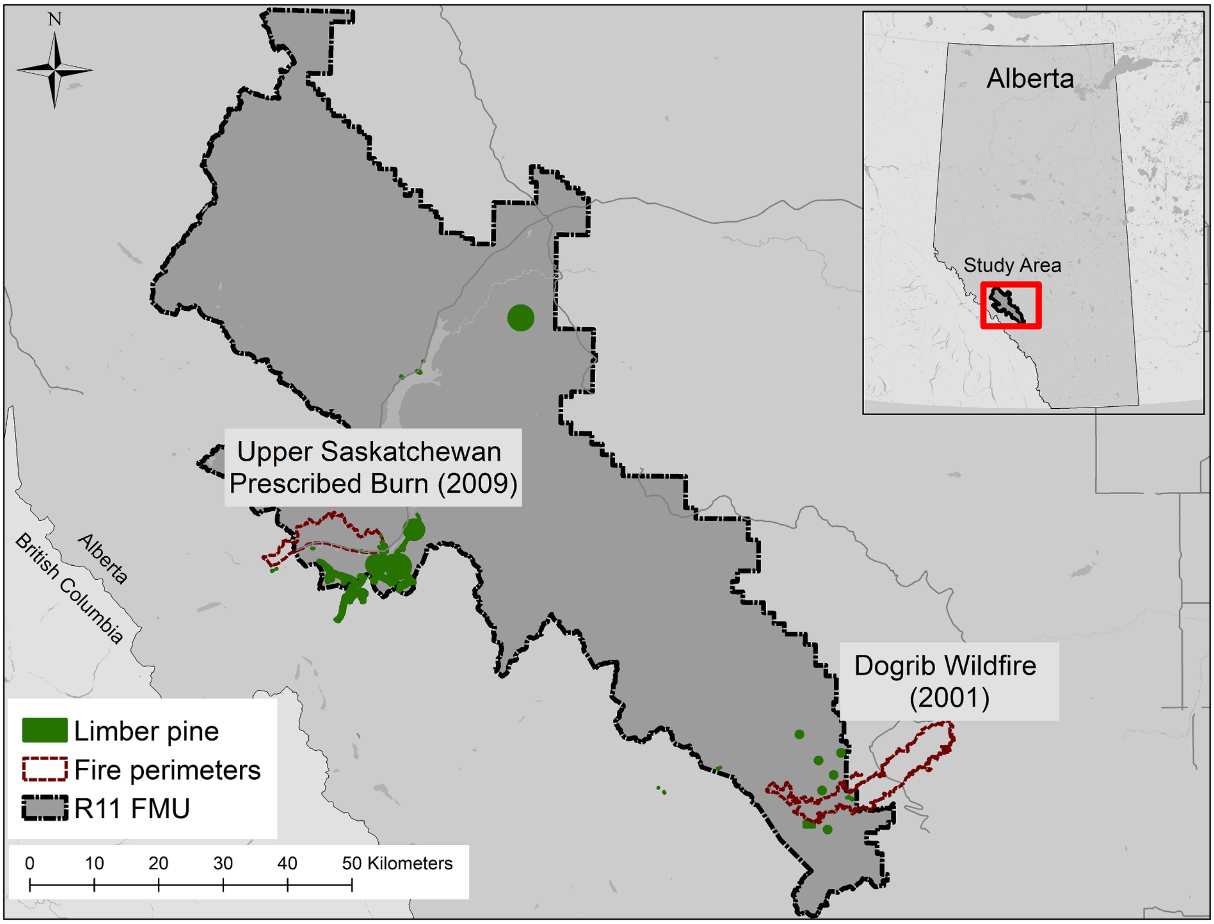

Fig. 1. Limber pine distribution in Alberta's R11 Forest Management Unit study area and the two fires surveyed in 2017. The Dogrib fire perimeter contained small limber pine seed sources, while the Upper Saskatchewan was not found to have substantive living limber pine within the fire perimeter.

representing two treatment blocks and one control, with abbreviations given in parentheses: limber pine present and dominant or co-dominant prior to fire (burned LP), limber pine absent or a very minor stand component prior to fire (burned absent), and unburned limber pine control (unburned) (Table 1). These stand types were chosen to determine if limber pine would re-colonize sites post-disturbance, if disturbance allowed limber pine to colonize new areas previously occupied by competitor species, and to establish a baseline of potential limber pine regeneration within each region.

We sampled within each stand type for limber pine regeneration and potential explanatory biophysical characteristics using a nested plot design. The three stand types mentioned above were subdivided into sites, with six sites per stand type dispersed throughout each region. Within each of the sites, four 50 × 4 m plots were established within generally homogeneous habitat for a total of 144 plots (Fig. 2). Sampling efforts were somewhat constrained by the number of suitable sites (specifically, burned LP stands) and the need to sample within a reasonable dispersal distance from limber pine seed sources. Plots were oriented perpendicular to the slope and typically placed along an elevation gradient, with distance between plots randomized between 25

Table 1
Topographic setting and stand structure of the stand types within each region. Values indicate the mean value at plots, with standard error shown in parentheses (n = 24 macro-plots within each fire and stand type category).

	Upper Saskatchewan (2009)			Dogrib (2001)		
Variable	Burned LP	Burned absent	Unburned	Burned LP	Burned absent	Unburned
Topography						
Elevation (m)	1556 (17)	1619 (20)	1426 (11)	1725 (23)	1697 (24)	1705 (9)
Heat load index	0.91 (0.003)	0.91 (0.004)	0.84 (0.019)	0.88 (0.016)	0.92 (0.004)	0.91 (0.005)
Slope (°)	18 (1.1)	23 (1.4)	27 (1.0)	30 (1.3)	28 (0.8)	31 (0.9)
Total basal area (m^2 ha^{-1})*						
Total**	5.2 (0.63)	5.6 (0.93)	10.4 (0.61)	5.1 (0.60)	6.6 (1.26)	7.5 (0.69)
Limber pine	2.2 (0.29)	0	8.6 (0.75)	1.8 (0.34)	0.2 (0.08)	4.2 (0.45)
White spruce	0.3 (0.12)	0.5 (0.21)	1.0 (0.25)	2.5 (0.62)	1.6 (0.33)	3.1 (0.42)
Lodgepole pine	2.7 (0.58)	4.3 (0.71)	0.67 (0.24)	0.6 (0.37)	4.0 (1.35)	0
Living limber pine basal area (m^2 ha^{-1})	0.27 (0.09)	0 (0)	6.97 (0.69)	0.10 (0.10)	0 (0)	3.58 (0.49)
Proportion overstory mortality	0.92 (0.14)	0.99 (0.02)	0.19 (0.17)	0.97 (0.09)	1.00 (0)	0.18 (0.12)

*Standing basal area of all living and dead trees above breast height.
**Unidentified dead basal area not reported separately but included in total.

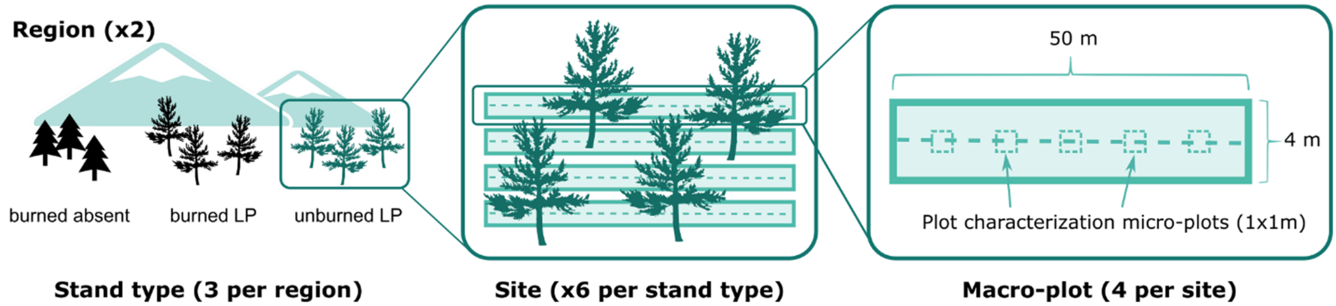

Fig. 2. Nested sampling design used in our study area. Biophysical explanatory variables measured in micro-plots were averaged to characterize the macro-plot. Macro-plots were used as the sampling unit and site was included as a random effect in all model analyses.

and 75 m. On a few occasions in which randomization placed plots in unsuitable habitat or in inaccessible topography, the plot was moved to the nearest suitable location or a new random distance was drawn. Data collected within each plot included elevation, aspect, and slope. Aspect was later translated to Heat Load Index in which zero represents the coolest slope and one the warmest, as in equation 3 of McCune and Keon (2002).

To characterize each large plot, we established five 1 m × 1 m micro-plots at 5, 15, 25, 35, and 45 m along the long axis of the macro-plot. Within each of these micro-plots, we measured ground cover of different vegetation classes (shrub, ground shrub, graminoids, and herb), deadwood, litter, rock, and exposed mineral soil. Soil texture and litter depth were measured by digging a small hole in the center of each micro-plot. Soil texture was classified using the flowchart from Archibald et al. (1996) and was later converted to numerical classes from 0 to 5, 0 being the coarsest, to allow averaging of the values for the plot (Table 2). Basal area was measured to species at each micro-plot using a basal area prism (factor 2), with separate tallies for living and dead trees of each species. In our fires, the majority of burned trees retained their main branches and a combination of identifying characteristics, namely, bark, retention of semi-serotinous cones (lodgepole) or absence of cones (typically consumed in Douglas fir, limber pine, and white spruce), and bole architecture (limber pine often in clusters with a unique, open silhouette). Dead trees were thus identified using their silhouette form, knowledge of the dominant species present at the site prior to the burn, and these other distinguishable clues. Unrecognizable trees and snags taller than breast height were placed in the category "unknown". Values from all five micro-plots were then averaged to arrive at an overall basal area, ground cover, canopy cover, and soil profile for each macro-plot.

A complete search for regeneration was conducted in the 50 × 4 m plot and seedlings (height < 1.4 m) of all tree species encountered were tallied. Caching by Clark's nutcrackers can sometimes result in several limber pine seedlings growing in a cluster (Carsey and Tomback, 1994). The number of seedlings within a cluster was noted and the height of each seedling recorded. For the purpose of statistical analysis, the number of limber pine regeneration sites, inclusive of both clustered and single individuals, was used as a representation of a single nutcracker caching event rather than number of stems, as in Leirfallom et al. (2015); we refer to these as "seedling clusters" (inclusive of both single stem and clustered individuals) throughout this document. Limber pine seedlings were aged by counting the annual growth whorls present on the stem (Daly and Shankman, 1985; Tomback et al., 1993). In cases of clustering, the tallest seedling was used to determine the age of the group. These age estimates were then used to differentiate limber pine regeneration that had germinated in the time since fire period from older seedlings that were advanced regeneration in both the burned and unburned plots. Although annual growth whorls tend to be a relatively

Table 2
Summary of predictor variables included in occurrence and abundance analyses of limber pine seedling establishment.

Model Category	Variable	Description
Seed	Seed distance*	Categorical variable indicating whether an available seed source is within 100 m of the macro-plot's center
	BA live LP	Basal area of alive limber pine averaged from all micro-plots
	Proportion LP	Proportion of standing limber pine (alive or dead) basal area
Microclimate	Elevation	Elevation at each macro-plot
	Heat load index	Transformed aspect, slope, and latitude representing typical heat load
	Slope	Slope at each macro-plot
	Canopy cover	Average canopy cover
Substrate	Texture	Soil texture converted to a numeric scale as below: 0 = Bedrock 1 = Rock, scree, gravel, etc. 2 = Loamy sand, sand, silty sand 3 = Loam, silt loam, sandy loam 4 = Silt 5 = Any clay
	Litter volume	Volume of fine dead material as measured by litter depth at micro-plot center multiplied by litter cover, averaged over the macro-plot
	Rock	Average of proportional cover measurements**
	Soil	
	Downed wood	
Vegetation	Other regen	Number of tree seedlings of other species (i.e., not limber pine) found in the macro-plot
	Ground shrub	Average of proportional cover measurements**
	Shrub	
	Herb	
	Grass	

*Used only in the occurrence analysis.
**Applies to rock, soil, downed wood, ground shrub, shrub, herb, grass.

accurate method of ageing, particularly for younger seedlings (Urza and Sibold, 2013), there is a possibility that the age group of seedlings whose true age was near the fire age may have been misclassified. Only seedlings from the post-fire age class were used in the analyses.

In order to map the distance between each plot and the nearest limber pine seed source, we conducted a helicopter survey of both the Upper Saskatchewan and Dogrib regions. This step was essential to address the challenge of missing potential seed sources, as the distribution of limber pine in Alberta is currently not thoroughly mapped. Areas with limber pine were spotted from the helicopter and marked on GPS for later investigation on the ground. Previously unmapped limber pine stands found in the Dogrib region were mapped by foot, using a Garmin 64S GPS to regularly mark stand boundary points, which were later converted to stand polygons in ArcGIS. No new limber pine stands were found in the Upper Saskatchewan; polygons of these stands had already been mapped and were obtained via the Alberta Conservation Information Management System (ACIMS). The distance from each plot's centroid to the nearest of these seed source stands was calculated in ArcGIS using planar distance via the "Generate Near Table" in the Analysis Tools, Proximity toolbox to arrive at the variable "seed distance".

2.3. Data analysis

All statistical analyses were performed in R version 3.4.3 (R Core Team, 2017) using generalized linear mixed-effects models (GLMMs) fit with the package glmmTMB (Brooks et al., 2017). Random intercepts were allowed by site to account for lack of independence between plots clustered within each site. Region was not included as a random intercept, as it had only two levels and random effect variables should have > 4 levels to allow calculation of inter-class variance (Zuur et al., 2013). Prior to fitting the models, all continuous explanatory variables were standardized so each had a mean of zero and standard deviation of one. Standardization simultaneously allowed variables at different scales to be more easily compared, while also improving model convergence (Zuur et al., 2009).

We first compared seedling densities between the three stand types (two burned and one unburned control) and two regions. We used a log-linked Poisson distribution to model counts of seedlings, using region and stand type as fixed-effects. Interactions between the two were assessed, though this was later discarded due to lack of significance. Size of the plots was used as an offset to convert seedling counts into a density.

We then performed two analyses to assess which of four proposed competing hypotheses of environmental/biotic factors had the strongest explanatory power for the observed patterns of regeneration occurrence and abundance. The four hypotheses (with abbreviations listed in parentheses) were: seed availability and distribution by Clark's nutcrackers (seed), abiotic substrate type (substrate), variables affecting microclimate that could act as a proxy for moisture availability and solar radiation (microclimate), and presence of competing vegetation (vegetation). Variables were assigned to hypotheses following exploratory data analysis as suggested in Zuur et al. (2010) to ensure no variables included within one hypothesis category had a variance inflation factor (VIF), a measure of multicollinearity severity, greater than 3 or correlation coefficient greater than 0.7 (Dormann et al., 2013; Zuur et al., 2010). The variables rock cover, litter depth, and litter cover were considered for the substrate hypothesis but found to exceed these thresholds; litter cover and litter depth were thus combined to create the variable "litter volume" which decreased its collinearity with rock cover to an acceptable level. The variable "seed distance" was included in the seed hypothesis for only the occurrence analysis; for reasons discussed below, the abundance analysis was limited to unburned stands which contained living mature limber pine that could act as an immediate seed source, making "seed distance" an unnecessary variable. "Seed distance" in the occurrence analysis was prone to quasi-complete separation due to near-perfect explanation of the response and was thus categorized to dampen this effect and allow its inclusion in the seed model (Zhao and Iyengar, 2010). A list of variables assigned to each hypothesis, including their description, are provided in Table 2.

Each hypothesis set was evaluated against the occurrence of limber pine seedlings in all stands and abundance of limber pine seedlings in unburned stands. The abundance analysis was conducted on only plots from the unburned stand type due to the low number of limber pine seedlings found in the burns. This restriction allowed greater exploration of variables important to limber pine regeneration when the hurdle of seed distribution distance was minimized and provided a simple paradigm for avoiding problems of zero-inflation (Zuur and Ieno, 2016). Occurrence models were fit using a binomial distribution to define the probability of seedling occurrence, given variables in each hypothesis. Abundance models were fit using a log-linked Poisson distribution and included an additional offset variable to account for the different times since fire in the Dogrib and Upper Saskatchewan regions. Hypotheses for the occurrence and abundance analyses were assessed using Akaike information criterion (AIC), following iterative removal of variables from each model category to ensure no hypothesis was penalized for complexity which did not contribute to an improved model fit (Burnham and Anderson, 2002). Region was considered as a fixed-effect within the global model of each hypothesis category for both the occurrence and abundance analyses but discarded, due to lack of significance and contribution to improving AIC, to further improve model parsimony and simplify the results. A combined model including all variables retained from each hypothesis category was then created to examine variable significance and the relative strength of each variable on seedling cluster occurrence and abundance, as judged by their coefficients, when used in combination to model the response.

Finally, variables retained in the top-performing abundance model were compared across stand types to examine how their characteristics differed between unburned and burned plots. This step was performed to determine if the relative absence of limber pine seedlings in the burn may be attributable to differences in these characteristics between stand types since the abundance analysis was only conducted on the plots from the unburned stand type. In these models, each variable indicated as important to abundance was taken as the response variable and modelled against region and stand type. Interactions between region and stand type were considered and retained when significant. Since many of these variables were bounded (i.e., proportional cover values ranging between 0 and 1), a beta-distributed GLMM was used, requiring a slight transformation to each environmental variable as suggested by Smithson and Verkuilen (2006) to remove all instances of zeros and ones.

3. Results

3.1. Seedlings found by stand type and region

Limber pine seedling cluster density within the two burned stand types was significantly lower than within the unburned stand type (Fig. 3, n = 144 macro-plots). No statistical difference was found between the burned absent and burned LP stand types. We found a total of six post-fire seedling clusters in the burned plots, five of those in the burned limber pine and one in the burned limber pine absent stand types. Four of these six seedlings clusters were from the Dogrib region; two were found in the Upper Saskatchewan. In comparison, one hundred twenty-four similarly aged seedling clusters were found in the unburned limber pine plots, 88 from the Dogrib region and 36 from the Upper Saskatchewan. Larger numbers of seedling clusters in the Dogrib region than the Upper Saskatchewan were to be expected, given different ages of seedlings considered for analysis in the two regions (8 years since fire in Upper Saskatchewan; 16 years since fire in Dogrib). The percentage of seedlings found in clusters rather than as single individuals within the unburned stand types was comparable; in Dogrib,

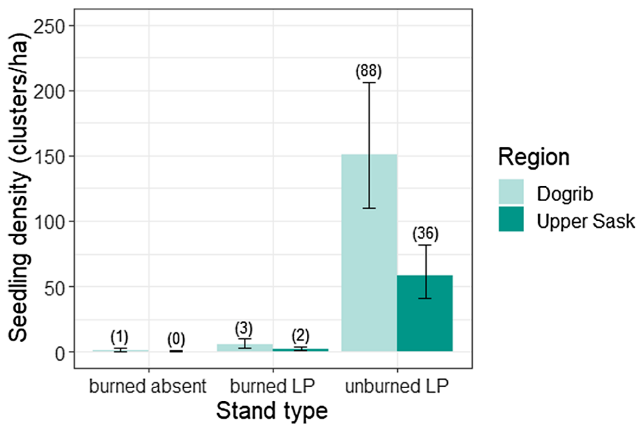

Fig. 3. Density of limber pine seedling clusters regenerating post-fire or in the post-fire age group in each stand type and region as modelled by Poisson GLMM. Error bars represent ± one standard error around each estimate; numbers above bars indicate the total number of seedling clusters observed.

26% of the seedlings found were in clusters, while in Upper Saskatchewan clusters comprised 28% of the seedlings. In the burned plots, one of the six regeneration sites found was in a cluster; this was found in the burned limber pine stand type in the Upper Saskatchewan. Although very few limber pine seedlings were found within the two burned stand types, seedlings of other tree species were present in burned plots surveyed. In both Upper Saskatchewan and Dogrib, fire appeared to boost regeneration of the dominant tree species in the region (lodgepole pine and white spruce, respectively), including in the stands where limber pine had been dominant/co-dominant prior to fire (burned LP stand type) (Fig. 4). In the burned LP plots in Dogrib, white spruce represented 81% of the post-fire regeneration, while in Upper Saskatchewan, 96% of post-fire regeneration in burned LP plots consisted of lodgepole pine.

3.2. Ecological processes influencing seedling occurrence

Of the hypotheses of ecological processes affecting seedling occurrence, the seed model was found to have the most support (Table 3; n = 144 macro-plots). Within this model, seed distance alone of the originally proposed variables was retained. Microclimate performed similarly well to the seed model (Table 3); only canopy cover was retained within this model. Substrate type and vegetation as hypotheses had negligible support, a trend which mostly continues when litter volume and downed wood (retained from the substrate hypothesis) and herb cover (retained from the vegetation hypothesis) were modelled in combination with canopy cover and seed distance (Table 3). Within this combined model, seed distance and canopy cover were found to have the greatest impact on the probability of seedling presence (Table 3). The log odds of post-fire limber pine seedling cluster presence declined by a factor of 5.566 at distances greater than 100 m from a seed source. It should be noted that thirteen percent of burned plots were sampled within 100 m of a seed source, while all the unburned plots were within this threshold distance. Canopy cover was associated with a greater likelihood of finding a seedling, with each unit of increase in canopy cover from the mean associated with a 1.557 increase in the log odds of seedling cluster presence.

3.3. Ecological processes influencing seedling abundance in unburned stands

Only the seed and substrate models were found to have support for driving seedling abundance in the unburned stands, with substrate found to have the definitively strongest support (Table 4; n = 48 macro-plots). Negligible support was found for the vegetation and microclimate hypotheses, suggesting that vegetative competition and microclimate do not play as strong a role as seed and substrate in limiting or promoting limber pine seedling recruitment within the unburned stands we surveyed where limber pine predominates (Table 4). From the substrate hypothesis, soil texture, rock cover, and exposed mineral soil cover were all selected as variables important to seedling abundance. Taken at the mean value of all other variables included in the model, limber pine seedling cluster abundance was positively influenced by rock cover and more fine soil textures, while negatively influenced by bare mineral soil cover (Table 4). In the seed model, basal area of alive limber pine was retained and found to have a positive association with seedling abundance (Table 4). From the less competitive vegetation and microclimate models, grass, number of other regenerating seedlings and canopy cover were retained and did not have a significant effect on seedling cluster abundance when analyzed in combination with the other variables.

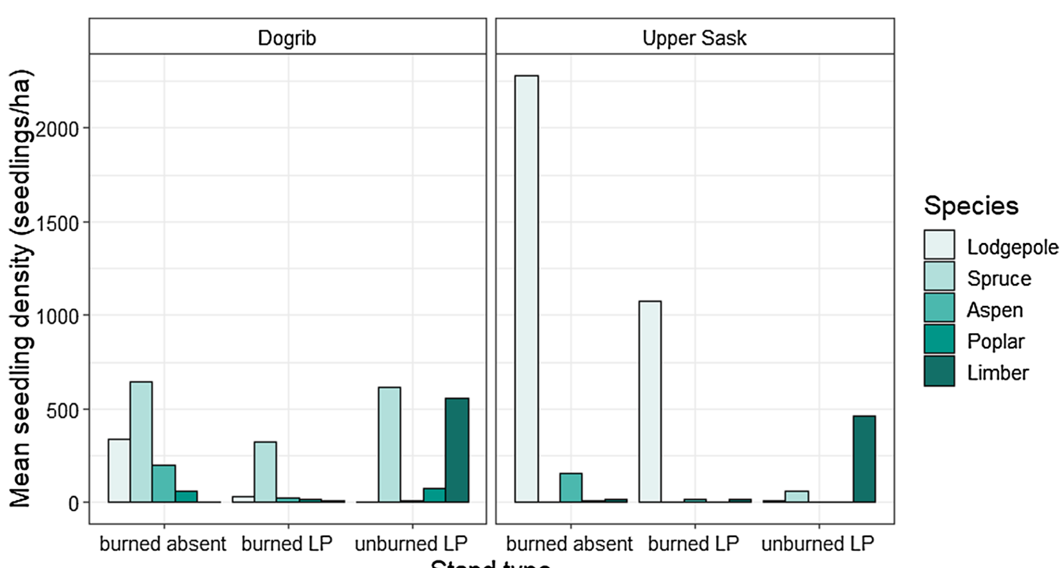

Fig. 4. Mean regeneration density of all species by stand type and region. Limber pine seedling cluster densities in the unburned stand type include both post-fire aged and older seedlings for comparison with other species, which were not aged.

Table 3
Comparison of ecological processes driving seedling occurrence, based on both AIC for most parsimonious models within each hypothesis category and comparison of effect size when all variables are modelled in combination. Effects in combined model standardized to facilitate comparison (n = 144 macro-plots).

		AIC comparison				Combined model		
Model category	Retained variables	K_i	AIC	ΔAIC	AIC_{wt}	β	SE	*P*
Seed	Seed distance	3	92.7	0	0.83	−5.566*	2.450	0.023
Microclimate	Canopy cover	3	95.9	3.2	0.17	1.557	0.776	0.045
Substrate	Litter volume	4	112.9	20.2	0.00	1.236	0.870	0.155
	Downed wood					−1.199	0.655	0.067
Vegetation	Herb	3	117.8	25.1	0.00	−1.214	0.833	0.145
Intercept	–	–	–	–	–	−0.895	1.044	0.391

*Factor = seed distance greater than 100 m.

3.4. Comparison of suitable substrates between stand types

Following the abundance analysis, all variables retained in the top hypothesis (substrate) were compared by region and stand type (n = 144 macro-plots). Because the abundance analysis was conducted on only unburned plots, this analysis was necessary for identifying whether factors important for limber pine recruitment in the unburned plots differed within the burned areas, thereby potentially negatively affecting limber pine recruitment following fire. Although some differences were found regionally between these variables, with significantly higher rock cover and lower soil cover in Dogrib than Upper Saskatchewan, they tended to be consistent between the three different stand types (Fig. 5). Specifically, soil cover and rock cover did not differ significantly between stand types (Table 5). Soil texture was found to be significantly finer in the burned limber pine absent stand type within the Dogrib region. As soil texture would not be affected by fire, this was an artifact of sampling design and in this region may simply demarcate the difference in where limber pine typically grows, as the unburned limber pine and burned limber pine stands had nearly identical soil texture values.

4. Discussion

4.1. Low limber pine recruitment in burned areas

Previous dendrochronological studies have described limber pine as having a competitive advantage over other tree species in colonizing areas post-fire, facilitated by long-distance dispersal by Clark's nutcrackers and the hardy, drought-resistance of limber pine seedlings (Rebertus et al., 1991). In the decade following fire, extirpated limber pine stands have also had documented recolonization by limber pine, due to the influence of Clark's nutcracker dispersal (Webster and Johnson, 2000). Our finding of limited post-disturbance recruitment of limber pine (5 seedlings in the burned LP stand type, one in the burned absent stand type) in the 8- and 16- years following fire provides an important, complementary examination of limber pine regeneration dynamics in the time period following disturbance, both under a changing climate and at the northernmost edge of its distribution. In contrast to previous findings, our survey of post-fire recruitment found immediate, and most abundant, regeneration not from limber pine as a post-disturbance colonist (Rebertus et al., 1991), but instead from the tree species which had been regionally dominant prior to fire (white spruce in Dogrib, lodgepole pine in Upper Saskatchewan) (Fig. 4). Seed dispersal from lodgepole pine aerial seedbanks and wind-dispersion from residual tree islands of white spruce (Greene and Johnson, 1999, 2000) resulted in post-fire recruitment 1–2 orders of magnitude greater than limber pine (Fig. 4). Our findings resonate with those of Coop and Schoettle (2009) who found that though limber pine was present in areas out to 100 m from burned edges, burning reduced the proportional abundance of limber pine due to fire-mediated boosts in recruitment of other species.

It is possible that low regeneration found within the burned areas surveyed could be due to insufficient time having passed for successful limber pine dispersal and germination to occur. In the Colorado Front Range, Coop and Schoettle (2009) did not perform their post-fire natural regeneration survey of limber pine until three decades following fire, while dendrochronological results of Donnegan and Rebertus (1999) found peak limber pine recruitment did not occur until 100 years post-fire. Despite this, Donnegan and Rebertus (1999) did find limber pine establishment within the first few years of their estimated time of disturbance. In addition, post-fire regeneration surveys of whitebark pine have found regeneration in the 5–18 years following extensive fire in Wyoming, Montana and Alberta (Tomback et al., 2001; Leirfallom et al., 2015; Salvador, 2018). Our survey for limber pine 8- and 16-years post-fire is within a similar post-disturbance time window as examined by other studies and an important description of limber pine regeneration dynamics in the immediate post-fire period. A recent study of seedling age distributions in mature limber pine stands suggests that recent seedlings (< 20 years old) comprise a high proportion of the total seedling population in regions with low white pine blister

Table 4
Comparison of ecological processes driving post-fire aged seedling abundance in the unburned stands, based on both AIC for most parsimonious models within each hypothesis category and comparison of effect size when all variables are modelled in combination. Effects in combined model standardized to facilitate comparison (n = 48 macro-plots).

		AIC comparison				Combined model		
Model category	Retained variables	K_i	AIC	ΔAIC	AIC_{wt}	β	SE	*P*
Substrate	Texture	5	196.7	0	0.98	0.477	0.188	0.011
	Soil					−0.607	0.187	0.001
	Rock					0.400	0.210	0.057
Seed	Basal area live limber pine	3	204.6	7.9	0.02	0.481	0.174	0.005
Vegetation	Grass	3	209.0	12.3	0.00	−0.288	0.164	0.079
	Other regen					0.007	0.189	0.967
Microclimate	Canopy cover	3	211.7	15.0	0.00	−0.142	0.158	0.369
Intercept	–	–	–	–	–	−1.912	0.216	< 0.001

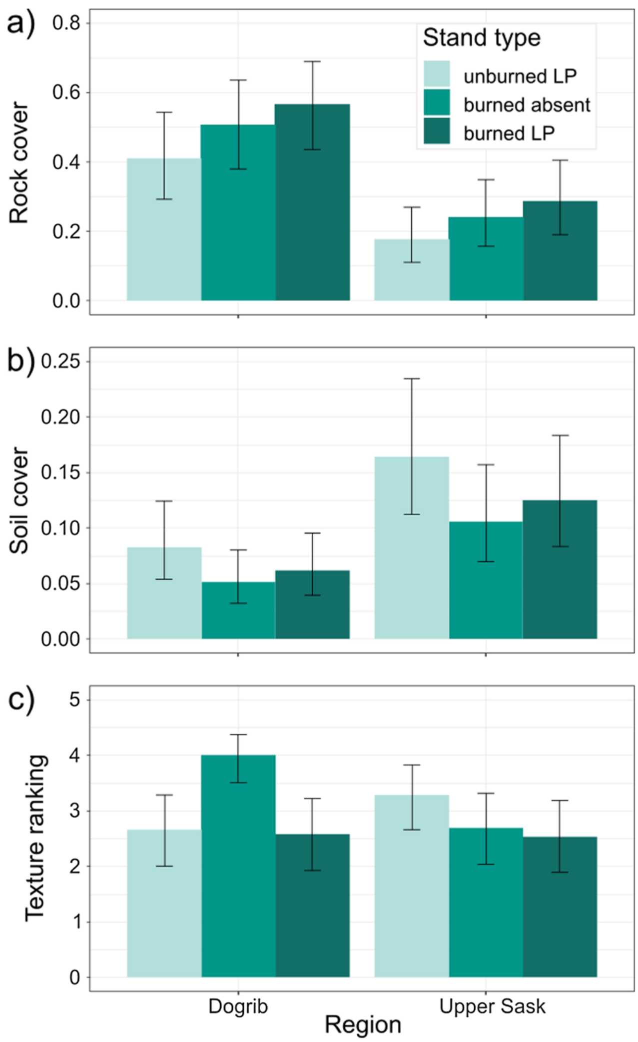

Fig. 5. Comparison of key driving variables for limber pine seedling abundance between stand types as estimated by beta-GLMMs for a) rock cover, b) soil cover, and c) texture (rankings provided in Table 2). Error bars represent confidence intervals. Estimates were back-transformed and displayed on the level of the variable being modelled to facilitate easy interpretation and comparison (n = 144 macro-plots).

Table 5
Comparison of relevant substrate variables by stand type and region. In all models, the unburned stand type and Dogrib region are taken as the first level of stand type and region. Interactions between region and stand type are shown when they were found to be significant. (n = 144 macro-plots).

Substrate variable	Effect	β	SE	*P*
Soil	Intercept	−2.408	0.234	< 0.001
	Burned absent	−0.508	0.283	0.073
	Burned LP	−0.317	0.281	0.259
	Region (Upper Sask)	0.780	0.232	< 0.001
Texture	Intercept	0.126	0.270	0.639
	Burned absent	1.271	0.385	0.001
	Burned LP	−0.061	0.381	0.873
	Region (Upper Sask)	0.530	0.382	0.165
	Burned absent × Region (Upper Sask)	−1.775	0.543	0.001
	Burned LP × Region (Upper Sask)	−0.563	0.539	0.296
Rock	Intercept	−0.357	0.269	0.184
	Burned absent	0.389	0.331	0.240
	Burned LP	0.628	0.331	0.057
	Region (Upper Sask)	−1.182	0.270	< 0.001

rust infections (Peters and Visscher, 2019). While a short period of time-since-fire may provide important regeneration opportunities, future regeneration surveys within the burned areas we surveyed may begin to find additional recruitment. It is important to note that recruitment for this species may occur on long-time scales and be sporadic in nature, depending on spatial and temporally varying factors such as post-disturbance climate, local nutcracker population behaviour, patterns of burn severity, and presence of competing species.

It is notable that limber pine regeneration within the unburned stands surveyed was found at healthy densities within the post-fire time period for both the Upper Saskatchewan and Dogrib regions (Fig. 3). Regeneration densities were comparable to or better than those found in other limber pine studies (Cleaver et al., 2016; Coop and Schoettle, 2009; Lanner and Vander Wall, 1980), especially after accounting for differences in time since disturbance and that other studies included all stems, rather than clusters, in their density measurements. The ~27% of seedlings found in clusters within the unburned stands was also within the range of 25–59% of clustering reported by Coop and Schoettle (2009), suggesting that nutcrackers were active in both regions within the post-fire time period, though it is important to note that singular seedlings can also arise from nutcracker caching activity (Tomback, 1982). It is unlikely that limitations to recruitment were due to seed shortages as mast years were observed directly in the Upper Saskatchewan region and more broadly across Alberta in 2010 (Peters et al., 2017), again in 2013 (VS Peters, personal communication), and are known to occur in limber pine every two to three years (Alberta Sustainable Resource Development and Alberta Conservation Association, 2007). It is thus likely that nutcrackers were active and had sufficient seed in these regions in the time-since-fire period and our observations of low regeneration within the burned areas was not caused by these factors.

4.2. Limitations to post-fire regeneration occurrence

Comparison of hypotheses for limber pine seedling occurrence revealed that distance to a seed source and canopy cover may have influenced the presence of limber pine seedlings across all stand types surveyed. In our analysis, use of quantitative seed distance was found to suffer from quasi-complete separation due to near-perfect explanation of seedling presence and absence, leading to inflated and biased parameter estimates (Zhao and Iyengar, 2010). To address this concern, seed distance was categorized by whether plots were found at greater or less than 100 m, to reflect the tendency of nutcrackers to cache within 100 m of a seed source stand (Hutchins and Lanner, 1982) and identify plots requiring longer distance seed distribution events. It should be noted that though this categorization was necessary, the majority of burned plots were sampled at distances greater than 100 m. As all the unburned plots were sampled at distances of less than 100 m, confounding effects may be present if conditions present in the burned sites precluded germination of seedlings.

Although nutcrackers have been observed caching whitebark pine seed at distances up to 32.6 km from a seed source stand (Lorenz et al., 2011), we found all limber pine seedlings within 250 m of a seed source. Much emphasis has been placed on long-distance dispersal events, especially as a basis for stocking of areas following disturbance. Several studies of nutcracker-mediated whitebark pine regeneration following fire, however, have found a strong, typically exponentially negative relationship between seed distance into a burn and seedling occurrence or abundance, though the maximum distance at which seedlings were found varied from a few hundred meters to several kilometres (Klutsch et al., 2015; Leirfallom et al., 2015; Moody, 2006; Tomback et al., 1990). Although long-distance caching certainly does occur, studies are finding that cache site selection by nutcrackers may be driven by proximity to overwintering habitat or nesting grounds, with longer distance dispersal occurring when seed sources and nutcracker home ranges do not overlap (Lorenz et al., 2011). Shorter distance caching is far more common (Tomback, 1978; Tomback and Linhart, 1990), potentially explaining why the extensively burned areas we surveyed only experienced limber pine recruitment in close proximity to seed sources.

It is possible, however, that nutcrackers were actively caching at further distances within the burn and that other factors associated with the burned areas precluded successful seedling germination. Seed pilferage by rodents may have had an effect, with possibly increased rates of predation in the burn rather than the unburned stands where greater seed availability could satiate predators (Nathan and Casagrandi, 2004; Orrock et al., 2006). Important mycorrhizal associates of limber pine may also have been less readily available in the burn, influencing the ability of seed cached within the burn to germinate and survive (Cripps and Antibus, 2011; Jenkins et al., 2018), though these conditions have not prevented post-fire establishment of five-needle pines within a similar time since fire period in other studies (Leirfallom et al., 2015; Tomback et al., 2001). Finally, recent studies have been finding reductions in forest resilience to disturbance under changing climate as warming temperature and increased annual moisture deficits become more common (Stevens-Rumann et al., 2017). Though climatic conditions appear to have not been severe enough to prevent establishment of competitor species in the burned areas we surveyed (Fig. 4) and limber pine's drought tolerance could provide an advantage under these conditions (Kueppers et al., 2017), it is possible that climatic factors had a greater effect on limber pine recruitment, given their location at the edge of their distributional range, than they did other species in the less-sheltered burned areas.

Our finding of the importance of microclimate, most notably of canopy cover, on limber pine seedling occurrence may have resulted from either nutcracker caching choice or seedling germination potential, as briefly described above. In the first case, it is possible that removal of overstory vegetation by fire may not have encouraged nutcracker caching in the context of our study area. Unburned stands surveyed in the two regions were quite open (maximum canopy cover = 33%); as such, there simply may not have been enough incentive, in terms of ease of cache retrieval due to earlier snowmelt in openings, for nutcrackers to travel outside of seed source stands on caching trips. In addition, although early observational studies frequently detected nutcrackers caching within open areas (Tomback, 1978; Vander Wall and Balda, 1977), a more recent radio-telemetry study found that caching was more common in close proximity to canopy cover, potentially as a concealment mechanism to reduce predation risk to nutcrackers (Lorenz et al., 2011). Nutcracker caching following fire thus may not be straightforward, instead occurring when the short-term exertion of long-distance travel is balanced by a longer-term energetic payoff due to ease of cache retrieval (McLane et al., 2017), which likely differs from fire event to fire event. Alternatively, though limber pine has been described as moderately shade-intolerant (Steele, 1990) and more open sky has been found to positively interact with limber pine recruitment (Coop and Schoettle, 2009), canopy cover offers greater thermoregulation and protection from direct solar radiation (Maher et al., 2005). In a seedling planting study, denser canopy cover improved limber pine seedling survival for the first four growing seasons, especially in cases where a protective nurse object was not available (Casper et al., 2016). Seedlings in burned areas with high overstory mortality (Table 1) may not have received establishment benefits from canopy cover (Gelderman et al., 2016), which was more present, albeit intermittent, within the unburned areas, potentially precluding seed that may have been cached in burned areas from successful germination.

4.3. Importance of seed and substrate to seedling abundance in unburned stands

Of the ecological processes hypothesized, substrate proved to have the strongest support in explaining patterns of seedling abundance in unburned stands, in which seed dispersal distance is less of a limiting factor. Of the variables tested, rock cover, soil texture, and mineral soil cover were retained. Though rocky sites tend to not represent the best growing conditions for most conifers, due to limited nutrient, soil, and water availability (Duryea and Dougherty, 1991), our results agree with those of other studies of limber pine regeneration which found a positive association between rock and limber pine recruitment (Cleaver et al., 2016; Coop and Schoettle, 2009; Peters and Visscher, 2019). Since our results describe the average conditions of the plot from which abundance measurements were taken, rocky conditions may have been representative of terrain conducive to limber pine's drought tolerance, providing it a competitive advantage in generally harsh conditions (Cleaver et al., 2016). Protruding rocks may also have been able to act as "nurse objects", facilitating seedling establishment (Casper et al., 2016). Alternatively, nutcrackers have previously been found preferentially caching in rocky areas (Tomback, 1982; Vander Wall and Balda, 1977), and so these findings may be indicative of nutcracker caching tendencies. Our finding of a negative association between regeneration and bare mineral soil cover is also consistent with previous studies (Cleaver et al., 2016; Coop and Schoettle, 2009); areas of open soil may indicate qualities of the site not conducive to plant success (Clever et al., 2016) or alternatively, areas in which seed caches are more readily retrieved by nutcrackers or predated by seed-eating rodents, as compared to rocky cavities (Munguía-Rosas and Sosa, 2008). Loose soil may also more easily erode or be washed away in these steep and windy habitats, potentially providing less stability for new seedlings to establish (Izlar, 2007). Soils of finer texture were found to promote limber pine recruitment. Soil texture at sites surveyed typically ranged from complete bedrock or gravel and shale to finer silts and silt loam; in this context, soil of finer texture will have greater moisture retention than sandier soils or gravel. Despite limber pine's renowned drought tolerance, greater levels of soil moisture will improve limber pine seedling survival and abundance (Moyes et al., 2013; Windmuller-Campione and Long, 2016) when this does not lead to being outcompeted by other species. Taken in combination with the finding of greater seedling abundance in rocky sites, our findings suggest that in-situ regeneration of limber pine is most greatly affected by availability of sites that strike a balance between moisture availability, rockiness, and limited open expanses of bare mineral soil.

Despite the greater level of importance of substrate in promoting limber pine recruitment in the unburned stands, comparison of our ecological process hypotheses showed that even when distance to a seed source is small, seed source dynamics still play an important role in regeneration of limber pine. Basal area of living limber pine proved to be an important factor in regeneration, especially in the combined model (Table 4). Other studies of limber pine regeneration echo the importance of proximal basal area of limber pine for limber pine regeneration as both a source of seed and shelter for limber pine seedlings (Cleaver et al., 2016; Peters and Visscher, 2019; Windmuller-Campione and Long, 2016). Nutcrackers are drawn to stands with a higher density of cones (Barringer et al., 2012); sites with denser limber pine may have a higher carrying capacity of nutcrackers and thus greater instances of seed caching and potential for regeneration, especially if nutcrackers are caching in close proximity to where seed was harvested. Nutcrackers have also been observed preferentially caching at the base of trees (Tomback, 1982), perhaps as a memory trigger, which may have influenced this relationship. Alternatively, areas of higher limber pine basal area may be indicative of higher overall site productivity and suitability for limber pine, thus also representing conditions most suitable to seedling germination and survival.

4.4. Similarity of substrates between burned and unburned plots

We found that burned and unburned plots were relatively spatially homogeneous with regards to substrates that were shown to be most important to limber pine in the unburned stands (i.e., bare mineral soil cover, rock cover, and soil texture). Conservation of plants benefits from identifying factors limiting recruitment, with historical studies often categorizing these factors as seed or substrate limitations (Clark et al., 1999; Eriksson and Erlhén, 1992). Post-disturbance environments tend to be seed-limited due to a combination of removal of vegetation (thereby creating an abundance of unoccupied desirable substrates) and reduction in readily available seed for species without a disturbance-resistant seed bank (Turnbull et al., 2000). By predominantly relying on distribution of seed by nutcrackers, rather than mechanisms employed by other species to colonize disturbed environments such as aerial seed banks and wind-dispersal, limber pine recruitment may be prone to seed-limitation. This is especially the case, as nutcracker dispersal behaviour is neither altruistic nor impartial, but rather, facilitates the later retrieval and consumption of cached seeds.

Interestingly, in the case of the fires we surveyed, burned versus unburned sites did not have different abundance of the substrates examined. Mature limber pine stands in Alberta are often found in open forests in which microsites remain available for continual recruitment of limber pine for hundreds of years (Webster and Johnson, 2000). Fire in these habitats may thus have less of a role in promoting recruitment than in stands in the more well-studied Colorado Front Range section of its distribution, in which fire-free intervals promote establishment of subalpine fir (*Abies lasiocarpa*) and eventual suppression of limber pine recruitment (Donnegan and Rebertus, 1999). Regional differences in limber pine fire ecology are to be expected, as within limber pine's broad geographic distribution it is found at both upper and lower treeline in grassy, open forests; exposed rocky slopes; and as a component of dense, mixed-conifer stands (Tomback and Achuff, 2010; Windmuller-Campione and Long, 2016). This spectrum of ecological contexts translates to an equally broad array of resulting fire regimes (Coop and Schoettle, 2011), co-occurring vegetation (Tomback and Achuff, 2010), climate, and likely differences in nutcracker caching behaviour (Lorenz et al., 2011), all influencing post-disturbance regeneration dynamics. Management of limber pine forests may need to fluctuate accordingly, with fire management in northernmost limber pine stands perhaps best served by focusing on protection from extensive, stand-replacing fires (Alberta Whitebark and Limber Pine Recovery Team, 2014), despite the role such disturbances may play in regeneration in other portions of its range (Donnegan and Rebertus, 1999).

5. Conclusions and management implications

We found that extensive, high-severity wildfire and prescribed burning did not stimulate rapid re-colonization and recruitment of limber pine in stands at the northernmost portion of its range. Occurrence of limber pine regeneration across both burned and unburned areas was best explained by variables related to seed dispersal and microclimate, while abundance of limber pine seedlings in the unburned areas was best explained by abiotic substrate values, most notably the soil texture and cover of rock and soil. Abundance of preferred substrates did not tend to differ by stand type, suggesting that suitable substrates may have been present within the burned areas surveyed. It is possible that limber pine stands at the margin of their range may be more sensitive to post-disturbance climatic conditions than stands elsewhere, thereby constraining the degree to which fire promotes limber pine recruitment. Our findings indicate that though limber pine has been found to have a competitive advantage in re-colonizing extensive, high-severity burns in portions of Colorado (Coop and Schoettle, 2009; Donnegan and Rebertus, 1999; Rebertus et al., 1991), the same may not hold true at the northernmost portion of its range.

Due to the lack of naturally occurring limber pine regeneration, the burned areas we surveyed in our study region would benefit from supplementary limber pine seedling plantings to achieve restoration goals. Wildfire events may provide opportunities to plant rust-resistant seedlings in substrate conducive to limber pine establishment, thereby capitalizing on an opportunity to increase population resilience to this often-lethal disease (Schoettle and Sniezko, 2007). Managers will note that areas under consideration for seedling plantings may require additional management activity, such as thinning of competitors like lodgepole pine recruits, which we found in abundance at some plots. Planted seedlings will benefit from considerations of substrate factors, including availability of nurse objects and competition-free microsites, in the selection of planting locations to ensure their success. Fire management in our study area may be best served by focusing on fire mitigation efforts to protect existing limber pine stands, due to the expense of planting new seedlings in stands extirpated by fire. Fuel treatments, such as thinning or prescribed burns like the Upper Saskatchewan fire, are advisable in areas adjacent to limber pine stands to reduce fire risk to highly valuable seed sources (Alberta Whitebark and Limber Pine Recovery Team, 2014; Keane, 2018), while providing the added benefit of reducing competition for limber pine.

Overall, our study provides much-needed insight into the post-fire regeneration dynamics of limber pine in the northern extent of its distribution. Given the range-wide differences in limber pine forest types across its diverse geographic and elevation range (Coop and Schoettle, 2011; Steele, 1990), fire management to promote the species will necessarily differ among regions, as evidenced by less immediate post-fire regeneration in our study than has been described in other portions of its range (Coop and Schoettle, 2009; Donnegan and Rebertus, 1999; Rebertus et al., 1991). The role of lower severity fire and microclimatic conditions post-fire in limber pine recruitment in our study area should be the focus of future research. We speculate that low-severity prescriptions may be of limited use in rocky, open stands such as those surveyed in this study, given the absence of fuel to carry such fire and the low degree of change to substrates even after stand-replacing fire. Future research could do well to focus on differences in fire management objectives across limber pine's distribution and different stand dynamics, as has been developed for whitebark pine (see Keane, 2018), to further advise managers on regional best-practices for the conservation of this important species.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Alberta Sustainable Resource Development and Alberta Conservation Association. 2007. Status of the limber pine (Pinus flexilis) in Alberta. Alberta Sustainable Resource Development, Wildlife Status Report No. 62. Edmonton, AB. 17 p.

Alberta Whitebark and Limber Pine Recovery Team, 2014. Alberta Limber Pine Recovery Plan 2014-2019. Alberta Environment and Sustainable Resource Development, Alberta Species at Risk Recovery Plan No. 35. Edmonton, AB. 61 p. https://open.alberta.ca/dataset/45ceb67a-7b44-4beb-a5a4-2b72a9b2e118/resource/05362a91-7f67-4af3-b8ea-e8e40f91a5dc/download/2014-sar-limberpine-recoveryplan-sep2014.pdf.

Archibald, J.H., Klappstein, G.D., Corns, I.G.W., 1996. Field Guide to Ecosites of Southwestern Alberta. Natural Resources Canada, Canadian Forest Service, Northwest Region, Northern Forestry Centre, Edmonton (AB), pp. 540.

Barringer, L.E., Tomback, D.F., Wunder, M.B., McKinney, S.T., 2012. Whitebark pine stand condition, tree abundance, and cone production as predictors of visitation by Clark's nutcracker. PLoS ONE 7 (5), e37663. https://doi.org/10.1371/journal.pone.0037663.

Baumeister, D., Callaway, R.M., 2006. Facilitation by *Pinus flexilis* during succession: a hierarchy of mechanisms benefits other plant species. Ecology 87 (7), 1816–1830.

Brooks, M.E., Kristenson, K., van Bentham, K.J., Magnusson, A., Berg, C.W., Nielsen, A., Skaug, H.J., Maechler, M., Bolker, B.M., 2017. glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed modeling. R J. 9 (2), 378–400.

Burnham, K.P., Anderson, D.R., 2002. Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach, 2nd ed. Springer, New York (NY), pp. 488.

Carsey, K.S., Tomback, D.F., 1994. Growth form distribution and genetic relationships in tree clusters of *Pinus flexilis*, a bird-dispersed pine. Oecologia 98 (3–4), 402–411. https://doi.org/10.1007/BF00324230.

Casper, A.M.A., Jacobi, W.R., Schoettle, A.W., Burns, K.S., 2016. Restoration planting options for limber pine (*Pinus flexilis* James) in the Southern Rocky Mountains. J. Torrey Bot. Soc. 143 (1), 21–37. https://doi.org/10.3159/TORREY-D-14-00085.

Clark, J.S., Beckage, B., Camill, P., Cleveland, B., Hillerislambers, J., Lichter, J., McLachlan, J., Mohan, J., Wyckoff, P., 1999. Interpreting recruitment limitation in forests. Am. J. Bot. 86 (1), 1–16.

Cleaver, C.M., Jacobi, W.R., Burns, K.S., Means, R.E., 2016. Limber pine regeneration and white pine blister rust in the central and southern Rocky Mountains. For. Sci. 63 (2), 151–164. https://doi.org/10.5849/forsci.16-052.

Coop, J.D., Schoettle, A.W., 2009. Regeneration of Rocky Mountain bristlecone pine (*Pinus aristata*) and limber pine (*Pinus flexilis*) three decades after stand-replacing fires. For. Ecol. Manage. 257 (3), 893–903. https://doi.org/10.1016/j.foreco.2008.10.034.

Coop, J.D., Schoettle, A.W., 2011. Fire and high-elevation, five-needle pine (Pinus aristata & P. flexilis) ecosystems in the southern Rocky Mountains: What do we know?, in: Keane, R.E., Tomback, D.F., Murray, M.P., Smith, C.M. (Eds.), The Future of High-Elevation, Five-Needle White Pines in Western North USDA Forest Service Proceedings RMRS-P-63. America: Proceedings of the High Five Symposium. 28–30 June 2010; Missoula, MT. USDA Forest Service Rocky Mountain Research Station, Fort Collins, CO: U.S., pp. 164–175.

COSEWIC, 2014. COSEWIC Assessment and Status Report on the Limber Pine *Pinus flexilis* in Canada. Committee on the Status of Endangered Wildlife in Canada, Ottawa https://wildlife-species.canada.ca/species-risk-registry/virtual_sara/files/cosewic/sr_Limber%20Pine_2014_e.pdf.

Cripps, C.L., Antibus, R.K., 2011. Native ectomycorrhizal fungi of limber and whitebark pine: Necessary for forest sustainability? in: Keane, R.E., Tomback, D.F., Murray, M.P., Smith, C.M. (Eds.), The Future of High-Elevation, Five-Needle White Pines in Western North USDA Forest Service Proceedings RMRS-P-63. America: Proceedings of the High Five Symposium. 28-30 June 2010; Missoula, MT. USDA Forest Service Rocky Mountain Research Station, Fort Collins, CO: U.S., pp. 37–44.

Cripps, C., Alger, G., Sissons, R., 2018. Designer niches promote seedling survival in forest restoration: a 7-year study of whitebark pine (*Pinus albicaulis*) seedlings in Waterton Lakes National Park. Forests 9, 477. https://doi.org/10.3390/f9080477.

Daly, C., Shankman, D., 1985. Seedling establishment by conifers above tree limit on Niwot Ridge, Front Range, Colorado, USA. Arct. Alp. Res. 17(4), 389–400. doi:10.1080/00040851.1985.12004046.

Donnegan, J.A., Rebertus, A.J., 1999. Rates and mechanisms of subalpine forest succession along an environmental gradient. Ecology 80 (4), 1370–1384. https://doi.org/10.1890/0012-9658(1999) 080[1370:RAMOSF]2.0.CO;2.

Dormann, C.F., Elith, J., Bacher, S., Buchmann, C., Carl, G., Carre, G., Garcia Marquez, J.R., Gruber, B., Lafourcade, B., Leitao, P.J., Munkemuller, T., McClean, C., Osborne, P.E., Reineking, B., Schroder, B., Skidmore, A.K., Zurell, D., Lautenbach, S., 2013. Collinearity: a review of methods to deal with it and a simulation study evaluating their performance. Ecography 36 (1), 27–46. https://doi.org/10.1111/j.1600-0587.2012.07348.x.

Duryea, M.L., Dougherty, P.M. (Eds.), 1991. Forest Regeneration Manual. Dordrecht, Springer, Netherlands, pp. 433.

Eriksson, O., Erlhén, J., 1992. Seed and microsite limitation of recruitment in plant populations. Oecologia 91 (3), 360–364. https://doi.org/10.1007/BF00317624.

Gelderman, M.S., Macdonald, S.E., Gould, A.J., 2016. Regeneration niche of whitebark pine in the Canadian Rocky Mountains: the basis to restoring an endangered species. Arct. Antarct. Alp. Res. 48 (2), 279–292. https://doi.org/10.1657/AAAR0015-057.

Goeking, S.A., Izlar, D.K., 2018. *Pinus albicaulis* Engelm. (whitebark pine) in mixed-species stands throughout its US range: broad-scale indicators of extent and recent decline. Forests 9, 131. https://doi.org/10.3390/f9030131.

Greene, D.F., Johnson, E.A., 2000. Tree recruitment from burn edges. Can. J. For. Res. 30 (8), 1264–1274. https://doi.org/10.1139/x00-040.

Greene, D.F., Johnson, E.A., 1999. Modelling recruitment of *Populus tremuloides*, *Pinus banksiana*, and *Picea mariana* following fire in the mixedwood boreal forest. Can. J. For. Res. 29 (4), 462–473. https://doi.org/10.1139/x98-211.

Hutchins, H.E., Lanner, R.M., 1982. The central role of Clark's nutcracker in the dispersal and establishment of whitebark pine. Oecologia 55 (2), 192–201. https://doi.org/10.1007/BF00384487.

Izlar, D.K., 2007. Assessment of whitebark pine seedling survival for Rocky Mountain plantings [thesis]. The University of Montana. 76 p.

Jenkins, M.L., Cripps, C.L., Gains-Germain, L., 2018. Scorched Earth: *Suillus* colonization of *Pinus albicaulis* seedlings planted in wildfire-impacted soil affects seedling biomass, foliar nutrient content, and isotope signatures. Plant Soil 425 (1–2), 113–131. https://doi.org/10.1007/s11104-018-3577-x.

Keane, R.E., 2018. Managing wildfire for whitebark pine ecosystem restoration in western North America. Forests 9, 648. https://doi.org/10.3390/f9100648.

Kinloch, B.B., 2003. White pine blister rust in North America: past and prognosis. Phytopathology 93, 1044–1047. https://doi.org/10.1094/PHYTO.2003.93.8.1044.

Klutsch, J.G., Goodrich, B.A., Jacobi, W.R., 2015. Post-fire regeneration dynamics in whitebark pine (*Pinus albicaulis*) forests in Wind River and Absaroka Mountains, Wyoming, USA. J. For. Res. 26 (3), 719–733. https://doi.org/10.1007/s11676-015-0064-6.

Kueppers, L.M., Faist, A., Ferrenberg, S., Castanha, C., Conlisk, E., Wolf, J., 2017. Lab and field warming similarly advance germination data and limit germination rate for high and low elevation provenances of two widespread subalpine conifers. Forests 8 (11), 433. https://doi.org/10.3390/f8110433.

Langor, D.W., 1989. Host effects on the phenology, development, and mortality of field populations of the mountain pine beetle, *Dendroctonus ponderosae* Hopkins (Coleoptera: Scolytidae). Can. Entomol. 121 (2), 149–157. https://doi.org/10.4039/Ent121149-2.

Lanner, R.M., Vander Wall, S.B., 1980. Dispersal of limber pine seed by Clark's nutcracker. J. For. 78 (10), 637–639. https://doi.org/10.1093/jof/78.10.637.

Leirfallom, S.B., Keane, R.E., Tomback, D.F., Dobrowski, S.Z., 2015. The effects of seed source health on whitebark pine (*Pinus albicaulis*) regeneration density after wildfire. Can. J. For. Res. 45 (11), 1597–1606. https://doi.org/10.1139/cjfr-2015-0043.

Lonergan, E.R., Cripps, C.L., Smith, C.M., 2014. Influence of site conditions, shelter objects, and ectomycorrhizal inoculation on the early survival of whitebark pine seedlings planted in Waterton Lakes National Park. For. Sci. 60 (3), 603–612. https://doi.org/10.5849/forsci.13-511.

Lorenz, T.J., Sullivan, K.A., Bakian, A.V., Aubry, C.A., 2011. Cache-site selection in Clark's nutcracker (*Nucifraga columbiana*). Auk 128 (2), 237–247. https://doi.org/10.1525/auk.2011.10101.

Maher, C.T., Nelson, C.R., Larson, A.J., Sala, A., 2018. Ecological effects and effectiveness of silvicultural restoration treatments in whitebark pine forests. For. Ecol. Manage. 429, 534–548. https://doi.org/10.1016/j.foreco.2018.07.040.

Maher, E.L., Germino, M.J., Hasselquist, N.J., 2005. Interactive effects of tree and herb cover on survivorship, physiology, and microclimate of conifer seedlings at the alpine tree-line ecotone. Can. J. For. Res. 35 (3), 567–574. https://doi.org/10.1139/X04-201.

McCune, B., Keon, D., 2002. Equations for potential annual direct incident radiation and heat load. J. Veg. Sci. 13 (4), 603–606. https://doi.org/10.1111/j.1654-1103.2002.tb02087.x.

McLane, A.J., Semeniuk, C., McDermid, G.J., Tomback, D.F., Lorenz, T., Marceau, D., 2017. Energetic behavioural-strategy prioritization of Clark's nutcrackers in whitebark pine communities: An agent-based modeling approach. Ecol. Modell. 354, 123–139. https://doi.org/10.1016/j.ecolmodel.2017.03.019.

Moody, R.J., 2006. Post-fire regeneration and survival of whitebark pine (Pinus albicaulis Engelm.) [thesis]. The University of British Columbia. 108 p.

Moyes, A.B., Castanha, C., Germino, M.J., Kueppers, L.M., 2013. Warming and the dependence of limber pine (*Pinus flexilis*) establishment on summer soil moisture within and above its current elevation range. Oecologia 171 (1), 271–282. https://doi.org/10.1007/s00442-012-2410-0.

Munguía-Rosas, M.A., Sosa, V.J., 2008. Nurse plants vs. nurse objects: effects of woody plants and rocky cavities on the recruitment of the *Pilosocereus leucocephalus* columnar cactus. Ann. Bot. 101, 175–185. https://doi.org/10.1093/aob/mcm302.

Nathan, R., Casagrandi, R., 2004. A simple mechanistic model of seed dispersal, predation and plant establishment: Janzen-Connell and beyond. J. Ecol. 92 (5), 733–746. https://doi.org/10.1111/j.0022-0477.2004.00914.x.

Orrock, J.L., Levey, D.J., Danielson, B.J., Damschen, E.I., 2006. Seed predation, not seed dispersal, explains the landscape-level abundance of an early-successional plant. J. Ecol. 94 (4), 838–845. https://doi.org/10.1111/j.1365-2745.2006.01125.x.

Perkins, J.L., 2015. Fire enhances whitebark pine seedling establishment, survival, and growth. Fire Ecol. 11 (2), 84–99. https://doi.org/10.4996/fireecology.1102084.

Peters, V.S., Gelderman, M.S., Visscher, D.R., 2017. Resiliency in masting systems: do evolved seed escape strategies benefit an endangered pine? Ecosphere 8 (9), e01928. https://doi.org/10.1002/ecs2.1928.

Peters, V.S., Visscher, D.R., 2019. Seed availability does not ensure regeneration in northern ecosystems of the endangered limber pine. Forests 10, 146. https://doi.org/10.3390/f10020146.

R Core Team. 2017. R: A language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria.

Rebertus, A.J., Burns, B.R., Veblen, T.T., 1991. Stand dynamics of *Pinus flexilis*-dominated subalpine forests in the Colorado Front Range. J. Veg. Sci. 2 (4), 445–458. https://doi.org/10.2307/3236026.

Retzlaff, M.L., Keane, R.E., Affleck, D.L., Hood, S.M., 2018. Growth response of whitebark pine (*Pinus albicaulis*) regeneration to thinning and prescribed burn release treatments. Forests 8, 311. https://doi.org/10.3390/f9060311.

Salvador, L.F.D., 2018. Fire and whitebark pine recovery strategies: drivers of post-fire natural regeneration. [thesis]. The University of Alberta. 116 p.

Schoettle, A.W., 2004. Ecological roles of five-needle pine in Colorado: Potential consequences of their loss, in: Sniezko, R.A., Samman, S., Schlarbaum, S.E., Kriebel, H.B. (Eds.), Breeding and Genetic Resources of Five-Needle Pines: Growth, Adaptability, and Pest Resistance RMRS-P-32. 23-27 Jul 2001; Medford, OR. USDA Forest Service, Rocky Mountain Research Station. pp. 124–135.

Schoettle, A.W., Sniezko, R.A., 2007. Proactive intervention to sustain high-elevation pine ecosystems threatened by white pine blister rust. J. For. Res. 12 (5), 327–336. https://doi.org/10.1007/s10310-007-0024-x.

Schoettle, A.W., Sniezko, R.A., Kegley, A., Burns, K.S., 2014. White pine blister rust resistance in limber pine: evidence for a major gene. Phytopathology 104 (2), 163–173. https://doi.org/10.1094/PHYTO-04-13-0092-R.

Smith, C.M., Langor, D.W., Myrholm, C., Weber, J., Gillies, C., Stuart-Smith, J., 2013. Changes in white pine blister rust infection and mortality in limber pine over time. Can. J. For. Res. 43 (10), 919–928. https://doi.org/10.1139/cjfr-2013-0072.

Smithson, M., Verkuilen, J., 2006. A better lemon squeezer? Maximum-likelihood regression with beta-distributed dependent variables. Psychol Methods. 11(1), 54–71. doi:10.1037/1082-989X.11.1.54.

Steele, R., 1990. Pinus flexilis James - limber pine, in: Burns, R.M., Honkala, B.H., (Eds.), Silvics of North America, Volume 1, Conifers. USDA Handbook 653. USDA Forest Service, Washington, DC, pp. 348–354.

Stevens-Rumann, C.S., Kemp, K.B., Higuera, P.E., Harvey, B.J., Rother, M.T., Donato, D.C., Morgan, P., Veblen, T.T., 2017. Evidence for declining forest resilience to wildfires under climate change. Ecol. Lett. 21 (2), 243–252. https://doi.org/10.1111/ele.12889.

Tomback, D.F., 1978. Foraging strategies of Clark's nutcracker. Living Bird 16, 123–161.

Tomback, D.F., 1982. Dispersal of whitebark pine seeds by Clark's nutcracker: a mutualism hypothesis. J. Anim. Ecol. 51 (2), 451–467. https://doi.org/10.2307/3976.

Tomback, D.F., Linhart, Y.B., 1990. The evolution of bird-dispersed pines. Evol. Ecol. 4 (3), 185–219.

Tomback, D.F., Sund, S.K., Hoffman, L.A., 1993. Post fire regeneration of *Pinus albicaulis*: height-age relationships, age structure, and microsite characteristics. Can. J. For. Res. 23, 113–119. https://doi.org/10.1139/x93-018.

Tomback, D.F., Anderies, A.J., Carsey, K.S., Powell, M.L., Mellmann-Brown, S., 2001. Delayed seed germination in whitebark pine and regeneration patterns following the Yellowstone fires. Ecology 82 (9), 2587–2600. https://doi.org/10.2307/2679938.

Tomback, D.F., Achuff, P., 2010. Blister rust and western forest biodiversity: ecology, values and outlook for white pines. For. Pathol. 40, 186–225. https://doi.org/10.1111/j.1439-0329.2010.00655.x.

Tomback, D.F., Sund, S.K., Hoffman, L.A., 1990. Coevolution of whitebark pine and nutcrackers: implications for forest regeneration. In: Schmidt W.C., McDonald K.J., (Eds). Symposium on whitebark pine ecosystems: Ecology and management of a high mountain resource. 29-31 March 1989; Bozeman, MT. Gen. Tech. Rep. INT-270. Ogden (UT): USDA Forest Service, Intermountain Research Station. p. 118–129.

Turnbull, L.A., Crawley, M.J., Rees, M., 2000. Are plant populations seed-limited? A review of seed sowing experiments. Oikos 88 (2), 225–238.

Urza, A.K., Sibold, J.S., 2013. Nondestructive aging of postfire seedlings for four conifer species in northwestern Montana. West J. Appl. For. 28 (1), 22–29. https://doi.org/10.5849/wjaf.11-014.

Vander Wall, S.B., Balda, R.P., 1977. Coadaptations of the Clark's nutcracker and the pinon pine for efficient seed harvest and dispersal. Ecol. Monogr. 47 (1), 89–111. https://doi.org/10.2307/1942225.

Webster, K.L., Johnson, E.A., 2000. The importance of regional dynamics in local populations of limber pine (*Pinus flexilis*). Écoscience 7 (2), 175–182. https://doi.org/10.1080/11956860.2000.11682586.

Windmuller-Campione, M.A., Long, J.N., 2016. Limber pine (*Pinus flexilis* James), a flexible generalist of forest communities in the intermountain west. PLoS One 11 (8), e0160324. https://doi.org/10.1371/journal.pone.0160324.

Zhao, M., Iyengar, S., 2010. Nonconvergence in logistic and poisson models for neural spiking. Neural Comput. 22 (5), 1231–1244. https://doi.org/10.1162/neco.2010.03-09-982.

Zuur, A.F., Ieno, E.N., Walker, N.J., Saveliev, A., Smith, G.M., 2009. Mixed Effects Models and Extensions in Ecology with R. Springer, New York, pp. 574.

Zuur, A.F., Ieno, E.N., Elphick, C.S., 2010. A protocol for data exploration to avoid common statistical problems. Methods Ecol. Evol. 1 (1), 3–14. https://doi.org/10.1111/j.2041-210X.2009.00001.x.

Zuur, A.F., Hilbe, J.M., Ieno, E.N., 2013. A Beginner's Guide to GLM and GLMM with R: A Frequentist and Bayesian Perspective for Ecologists. Highland Statistics, Newburgh, UK, pp. 253.

Zuur, A.F., Ieno, E.N., 2016. A Beginner's Guide to Zero-Inflated Models with R. Highland Statistics Ltd, Newburgh (UK), pp. 414.